

L11 — Animation II

Data-driven animation and speech-driven faces

Study summary generated from lecture slides. ETH AI for Digital Characters.

Why ML for animation?

- MoCap cannot cover all possible chatbot responses
- Procedural methods are repetitive; need on-the-fly synthesis
- Active listening: generate plausible non-speaker animations with context

Datasets

- BIWI 3D, MEAD, VOCASET: 3D face scans with emotion
- IEMOCAP: dyadic dialogs, multi-modal, 9 emotions + VAD
- BEAT, Talking with Hands: co-speech gestures

Learning-based methods

- Holden's methods: phase-functioned neural networks for locomotion
- DeepPhase: phase-aware autoencoder for periodic motion manifolds
- Normalizing flows: invertible maps, exact likelihood via Jacobian determinant
- Applications: diverse motion generation with temporal consistency

Speech-driven animation

- Procedural: amplitude from audio waveform
- FaceFormer: transformer for speech-driven 3D facial animation
- Blendshapes: brow, eyes, jaw, mouth controls

Evaluation

- FID: realism of generated motion distribution
- Foot sliding / contact error for locomotion